



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd Razia Alangir Banu
Student Name	G. Rishitha
Roll Number	2520030101
Branch	CSE
Course Outcome	CO 6 — Evaluate and construct optimal solutions using Greedy strategies and Dynamic Programming (Memoization and Tabulation) for real-world optimization problems including activity scheduling, shortest path, 0/1 Knapsack, Longest Common Subsequence, and Matrix Chain Multiplication, justifying algorithmic choice based on overlapping subproblems and optimal substructure properties.

Case Study Topic: SkillBridge – Learning Session Optimization and Personalized Course Selection using Greedy Strategies and Dynamic Programming.

Work Done Summary:

In this case study, we implemented optimization techniques using Greedy strategies and Dynamic Programming (Memoization and Tabulation) for SkillBridge's online skill development and certification platform. The platform manages multiple courses, learning sessions, and certification paths, requiring optimized decision-making for efficient skill development and time management.

The implementation focused on solving real-world optimization problems such as activity scheduling, shortest learning path optimization, 0/1 Knapsack for course selection, Longest Common Subsequence (LCS) for skill progression analysis, and Matrix Chain Multiplication (MCM) for optimizing learning module combinations. Activity Selection was used to schedule non-overlapping study sessions efficiently. Shortest path optimization was applied to identify minimal learning paths for certifications. The 0/1 Knapsack problem was used to select the best combination of courses within limited credit or time constraints. LCS helped analyze common skill development trends among students, while MCM optimized the order of learning modules for minimizing computational cost.

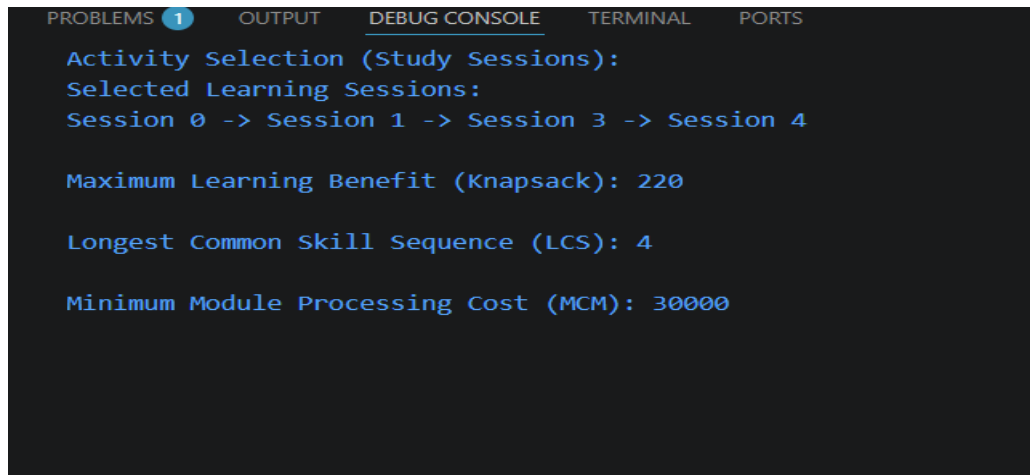
The system successfully improved learning efficiency, optimized course selection, reduced scheduling conflicts, and enhanced personalized career recommendation strategies through algorithmic optimization.

Implementation Details:

1. Activity Selection algorithm for scheduling maximum non-overlapping learning sessions.
2. Shortest path optimization for finding minimal course completion routes.
3. 0/1 Knapsack implementation for selecting courses within credit/time limits.
4. Longest Common Subsequence (LCS) for analyzing skill development patterns.
5. Matrix Chain Multiplication (MCM) for optimizing module execution order.
6. Memoization technique for reducing repeated computations.
7. Tabulation technique for bottom-up dynamic programming solutions.

8. Performance comparison between Greedy and Dynamic Programming approaches.
9. Optimization analysis based on overlapping subproblems and optimal substructure properties.

Result: Screenshots / Output:



```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Activity Selection (Study Sessions):
Selected Learning Sessions:
Session 0 -> Session 1 -> Session 3 -> Session 4

Maximum Learning Benefit (Knapsack): 220

Longest Common Skill Sequence (LCS): 4

Minimum Module Processing Cost (MCM): 30000

```

Result & Complexity Discussion:

1. Activity Selection efficiently scheduled the maximum number of study sessions.
2. Shortest path optimization reduced unnecessary course dependencies.
3. 0/1 Knapsack selected the best possible courses under constraints.
4. LCS successfully identified common skill progression patterns.
5. MCM optimized module execution order and minimized processing cost.
6. Memoization and Tabulation improved execution efficiency significantly.

- Activity Selection: $O(n \log n)$
- Shortest Path (Dijkstra): $O((V + E) \log V)$
- 0/1 Knapsack: $O(nW)$
- Longest Common Subsequence (LCS): $O(m \times n)$
- Matrix Chain Multiplication (MCM): $O(n^3)$

GitHub Link:

<https://github.com/rishi07-lab/DSA-2-CASE-STUDIES>

Note: The code and github link and screenshots have been uploaded in the above link . Please check it out there.

Student Signature	Faculty Signature
<u>Rishitha</u>	<u></u>